Structure-preserving time integrators

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Differential equations have been long studied and classified by their various structural properties. Examples include dissipative systems such as gradient flows, conservative systems with Hamiltonian structure, etc. Some problems are defined on structured spaces, e.g., differential equations on Lie groups. When structure is present, there are often advantages to preserving that structure in the numerical method. In some cases (e.g., the Lie group setting), the solution only makes sense if it lies on the group manifold; in other cases, the structure may underly important physics (conservation of mass or energy). When we discuss Monte Carlo methods, we will also see examples where the mathematical properties of a gradient or Hamiltonian flow simplify the conditions needed to ensure unbiased sampling. In these notes we discuss numerical integration methods from the point of view of preserving the underlying structure of the differential equation.

1 Structural properties

In this section we review a number of structural properties of interest.

We consider an autonomous differential equation

$$\dot{x} = f(x), \qquad x(t) \in \mathbf{R}^d, \quad f : \mathbf{R}^d \to \mathbf{R}^d. \tag{1}$$

The solution to the differential equation is given by the flow $\phi_\tau : \mathbf{R}^d \to \mathbf{R}^d$, such that $x(t+\tau) = \phi_\tau(x(t))$. If there is a set $\mathcal{D} \subset \mathbf{R}^d$ such that $\phi_\tau : \mathcal{D} \to \mathcal{D}$ for all $\tau \in \mathbf{R}$, we say that the pair $(\mathcal{D}, \phi_\tau)$ define a *dynamical system.* The flow defines a one-parameter group with respect to τ:

$$\rho_\tau \circ \rho_\sigma = \rho_{\tau+\sigma} = \rho_\sigma \circ \rho_\tau. \tag{2}$$

In particular, $\rho_\tau^{-1} = \rho_{-\tau}$.

1.1 Time-reversal symmetries

A differential equation (1) is said to be *ρ-reversible* if there exists an invertible matrix $\rho \in \mathbf{R}^{d\times d}$ such that $f(\rho x) = -\rho f(x)$, for all $x \in \mathcal{D}$. In this case, the flow satisfies

$$\rho \circ \phi_\tau = \phi_{-\tau} \circ \rho.$$

Example. The Lotka-Volterra equations (with all coefficients equal to unity)

$$\dot{u} = uv - u, \qquad \dot{v} = v - uv$$

are ρ-reversible with $\rho = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

A large class of mechanical systems are described by Newton's equations

$$M\dot{q} = p, \qquad \dot{p} = -\nabla V(q),$$

where, as we will see later, $q(t) \in \mathbf{R}^d$ is the vector of generalized coordinates specifying the configuration of the system, $p(t) \in \mathbf{R}^d$ are the associated generalized momenta, $M \in \mathbf{R}^{d\times d}$ is a positive definite (typically diagonal) mass matrix, and $V : \mathbf{R}^d \to \mathbf{R}$ is the potential energy. All such systems are ρ-reversible under the transformation

$$\rho(q,p) = (q,-p), \qquad \rho = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}.$$

The physical interpretation of this symmetry is that if we propagate the system forward in time, then change the sign of the velocity, the system will retrace its path.

1.2 First integrals

A function $C : \mathcal{D} \to \mathbf{R}$ is a first integral of (1) if $C \circ \phi_\tau(x) = C(x)$ for all $x \in \mathcal{D}$ and all $\tau \in \mathbf{R}$. In this case, every orbit of (1) lives on a level set of the function C. If C is differentiable, then we find that

$$0 = \frac{d}{dt}C(x(t)) = \nabla C \cdot \frac{dx}{dt} = \nabla C \cdot f(x), \qquad \forall x \in \mathcal{D}.$$

Physically, first integrals often represent important conservation laws such as conservation of mass or conservation of energy.

Linear conservation laws take the form $C(x) = b^T x$ for some constant vector $b \in \mathbf{R}^d$. The vector field satisfies $b^T f(x) = 0$ for all $x \in \mathbf{R}^d$. Solutions lie on the hyperplanes $b^T x(t) = b^T x(0) = const.$

Quadratic conservation laws take the form $C(x) = \frac{1}{2}x^T Ax + b^T x$, where $A = A^T$ is a constant symmetric matrix and $b \in \mathbf{R}^d$ a constant vector. In this case, the vector field satisfies $(Ax + b)^T f(x) = 0$ for all x. If A is positive definite, then the level sets of $C(x)$ are ellipsoids, and their preservation ensures boundedness of all solutions.

Example. The basic SIR model for spread of a communicable disease is

$$\begin{aligned} \dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \gamma I \\ \dot{R} &= \gamma I. \end{aligned}$$

Here, $S(t)$, $I(t)$ and $R(t)$ denote the sizes of the population that are susceptible, infectious and recovered, respectively, where the model assumes the disease is nonfatal and can only be contracted once. This SIR model has a linear conservation law with $b = (1,1,1)^T$, representing the conservation of the total population.

1.3 Hamiltonian systems

A Hamiltonian system is one of the form

$$\dot{x} = J\nabla H(x), \tag{3}$$

where $H(x) : \mathcal{D} \to \mathbf{R}$ is the Hamiltonian function and $J = -J^T$ is a constant, nonsingular (i.e. invertible), skew-symmetric matrix. Since any skew-symmetric matrix satisfies

$$x^T Jx = 0, \quad \forall x,$$

it follows that the Hamiltonian H is a first integral of the Hamiltonian system:

$$\frac{dH}{dt} = \nabla H \cdot \frac{dx}{dt} = \nabla H \cdot J\nabla H = 0$$

Hamiltonian systems have much more structure however. The nonsingularity requirement of J implies the dimension d must be even.

A matrix $F \in \mathbf{R}^{d\times d}$ is said to be a symplectic matrix (with respect to J) if

$$F^T J^{-1} F = J^{-1}.$$

A transformation $\Phi(x) : \mathcal{D} \to \mathcal{D}$ is said to be *symplectic transformation* if its Jacobian matrix is a symplectic matrix,

$$D\Phi(x)^T J^{-1} D\Phi(x) = J^{-1}, \quad \forall x \in \mathcal{D}.$$

The flow of a Hamiltonian system is a symplectic transformation. The matrix of partial derivatives of ϕ_t with respect to the initial data is $D\phi_t(x) = \frac{\partial}{\partial x_0}\phi_t(x_0)$. The flow $\phi_t(x_0)$ satisfies the differential equation

$$\frac{\partial}{\partial t}\phi_t(x_0) = J\nabla H(\phi_t(x_0)).$$

Taking the derivative with respect to the initial data yields the matrix differential equation

$$\frac{\partial}{\partial t} D\phi_t(x_0) = JH_{xx}(\phi_t(x_0))D\phi_t(x_0),$$

where H_{xx} denotes the Hessian matrix of H. Consequently, we note that

$$\begin{aligned}
\frac{d}{dt}\left(D\phi_t^T J^{-1} D\Phi_t\right) &= \left(\frac{\partial}{\partial t} D\phi_t\right)^T J^{-1} D\Phi_t + (D\phi_t)^T J^{-1} \frac{\partial}{\partial t} D\phi_t \\
&= (D\phi_t)^T H_{xx}^T J^T J^{-1} D\Phi_t + (D\phi_t)^T J^{-1} J H_{xx} D\phi_t \\
&= -(D\phi_t)^T H_{xx}^T D\Phi_t + (D\phi_t)^T H_{xx} D\phi_t \\
&= 0,
\end{aligned}$$

due to the symmetry of the Hessian matrix. We conclude that the matrix function $D\phi_t^T J^{-1} D\Phi_t$ is a constant of the flow. On the other hand, the flow satisfies $\phi_0(x_0) = x_0$, and therefore $D\phi_0 = I$, the identity matrix. Therefore, we find

$$D\phi_t^T J^{-1} D\Phi_t = D\phi_0^T J^{-1} D\Phi_0 = J^{-1},$$

proving the assertion.

A final conclusion can be draw by computing the determinant of both sides of the above formula, which yields

$$\det(D\phi_t)^2 \det J^{-1} = \det J^{-1},$$

from which it follows that $\det(D\phi_t) = 1$, implying the flow of a Hamiltonian system preserves the Lebesgue measure. (The possibility $\det(D\phi_t) = -1$ is precluded by the earlier observation that $D\phi_0 = I$ and continuity of the flow.)

2 Adjoint methods and composition

Let a numerical method be given by the map

$$x_{n+1} = \psi_{\Delta t}(x_n), \qquad \psi_{\Delta t} : \mathcal{D} \to \mathcal{D}.$$

The adjoint map is given by

$$\psi^*_{\Delta t} = \psi^{-1}_{-\Delta t},$$

that is, the adjoint is the inverse of the method with negative time step. To construct the adjoint we reverse the sign of Δt, swap x_{n+1} and x_n, then view the result as a method for computing x_{n+1}. For example, to compute the adjoint of Euler's method we

$$x_{n+1} = x_n + \Delta t f(x_n) \qquad \mapsto \qquad x_n = x_{n+1} - \Delta t f(x_{n+1}),$$

which can be rewritten as backward Euler

$$x_{n+1} = x_n + \Delta t f(x_{n+1}).$$

The adjoint of forward Euler is backward Euler and vice versa. The adjoint of an explicit Runge-Kutta method is again a Runge-Kutta method. The adjoint of an explicit Runge-Kutta method is always implicit (but the adjoints of most implicit Runge-Kutta method are also implicit). If $\psi^*_{\Delta t} = \psi_{\Delta t}$ the method is self-adjoint. Two examples of self-adjoint methods are the trapezoidal rule

$$\frac{x_{n+1} - x_n}{\Delta t} = \frac{1}{2}\left(f(x_{n+1}) + f(x_n)\right)$$

and the implicit midpoint rule

$$\frac{x_{n+1} - x_n}{\Delta t} = f\left(\frac{x_{n+1} + x_n}{2}\right). \tag{4}$$

Due to cancellation of terms, the Taylor series expansions of the (local) truncation error of self-adjoint methods retain only odd powers of Δt. Consequently, self-adjoint method always have global error of even order. If f satisfies a ρ-reversal symmetry, then so does a self-adjoint Runge-Kutta method. Such methods are also called *reversible*.

Given two numerical methods $\psi_{1,\Delta t}$ and $\psi_{2,\Delta t}$, we can create a new method by composition

$$\bar{\psi}_{\Delta t} = \psi_{1,\alpha\Delta t} \circ \psi_{2,(1-\alpha)\Delta t}.$$

A method composed this way will typically retain the order of the component method of lower order. However, the composition of a method with its adjoint yields a self-adjoint method, which must have even order. For example, composing the first order forward and backward Euler methods (each first order) yield the trapezoidal rule and implicit midpoint rule, depending on the order of composition, and because these are self-adjoint, they are of order two.

The maps of numerical methods typically do not satisfy the group property (2). Consequently, composing a method with itself yields a nontrivial result. In particular, we can choose a set of coefficients $\alpha_1 + \cdots + \alpha_s = 1$, and consider the self-composition method

$$\bar{\psi}_{\Delta t} = \psi_{\alpha_1\Delta t} \circ \cdots \circ \psi_{\alpha_s\Delta t}.$$

If the component method has order p, then a Taylor series expansion shows that the above method has order $p+1$ if the coefficients satisfy the condition

$$\alpha_1^{p+1} + \cdots + \alpha_s^{p+1} = 0.$$

The above equation has no solution for odd p. However, for even p a three-term solution exits

$$\alpha_1 = \alpha_3 = \frac{1}{2 - 2^{\frac{1}{p+1}}}, \qquad \alpha_2 = \frac{-2^{\frac{1}{p+1}}}{2 - 2^{\frac{1}{p+1}}}.$$

If $\psi_{\Delta t}$ is self-adjoint, then

$$\bar{\psi}_{\Delta t} = \psi_{\alpha_1\Delta t} \circ \psi_{\alpha_2\Delta t} \circ \psi_{\alpha_3\Delta t}$$

is again self-adjoint (in this case, we call the method the Yoshida composition). Its order is at least $p+1$, but because it is self-adjoint, its order must be even, consequently at least $p+2$. Applying this method recursively, we can construct methods of arbitrarily high order given any self-adjoint method.

3 Splitting methods

Sometimes (actually, surprisingly often) we can decompose a vector field into the sum of exactly integrable vector fields, such that the differential equation (1) can be written

$$\dot{x} = f_1(x) + \cdots + f_s(x),$$

where we know the exact flow of $\dot{x} = f_i(x)$, for $i = 1, \ldots, s$. let $\phi_{i,t}$ denote the exact flow of the vector field f_i. Then the composition

$$\psi_{\Delta t} = \phi_{1,\Delta t} \circ \cdots \circ \phi_{s,\Delta t}$$

is a numerical method of order at least $\mathcal{O}(\Delta t)$. Composing this flow with its adjoint

$$\psi^*_{\Delta t} = \phi_{s,\Delta t} \circ \cdots \circ \phi_{1,\Delta t}$$

yields a self-adjoint method, which must be of order at least $\mathcal{O}(\Delta t^2)$. Subsequently, we can use the results of the previous section to extend the method to arbitrarily high orders.

If $f(x)$ satisfies a ρ-reversal symmetry, and also each of the split vector fields f_i satisfies the same symmetry, then the self-adjoint splitting method is also ρ-reversible.

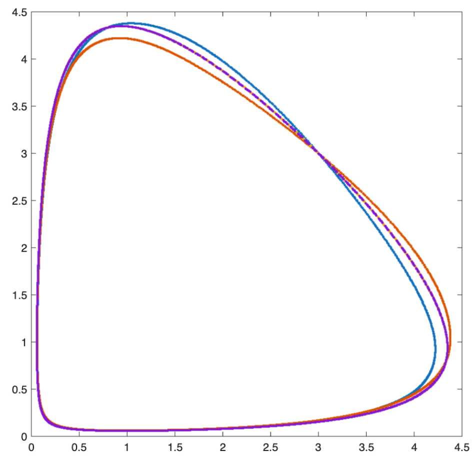
Example. Consider two spltting methods for the Lotka-Volterra equation. The first splitting is based on the vector fields

$$f_1 = \begin{pmatrix} uv - u \\ 0 \end{pmatrix}, \qquad f_2 = \begin{pmatrix} 0 \\ v - uv \end{pmatrix},$$

which do not preserve the ρ-reversal symmetry. The second splitting is base on the vector fields

$$f_1 = \begin{pmatrix} uv \\ -uv \end{pmatrix}, \qquad f_2 = \begin{pmatrix} -u \\ v \end{pmatrix}$$

which does preserve the ρ-reversal symmetry. All vector fields are exactly integrable. In the following figure we solve the Lotka-Volterra equations with the symmetric, second order splittings based on these vector fields. We plot the phase plots of $(u(t), v(t))$ and $(v(t), u(t))$ on the same axes using a large time step ($\Delta t = 0.3$). The curves for the reversible splitting are identical, for the irreversible splitting they are not so.



(In this figure, the red and blue lines represent $(u(t), v(t))$ and $(v(t), u(t))$ for the irreversible splitting. The curves for the reversible splitting are yellow and purple.)

4 Conservation of first integrals

A numerical method conserves a first integral C if $C \circ \psi_{\Delta t}(x) = C(x)$ for all $x \in \mathcal{D}$. It is straightforward to see that a generic s-stage Runge-Kutta method,

$$F^{(i)} = f(x_n + \Delta t \sum_{j=1}^{s} a_{ij} F^{(j)}), \quad i = 1, \ldots, s,$$
$$x_{n+1} = x_n + \Delta t \sum_{j=1}^{s} b_j F^{(j)},$$

conserves arbitrary linear first integrals $C(x) = b^T x$. Using the condition $\nabla C \cdot f = b^T f(x) = 0$, we multiply all equations above from the left with b^T to find

$$b^T F^{(i)} = 0, \quad i = 1, \ldots, s, \qquad b^T x_{n+1} = b^T x_n,$$

hence $C(x_{n+1}) = C(x_n)$.

For quadratic first integrals the situation is significantly less generic. Only the family of Gauss-Legendre Runge-Kutta (GLRK) methods (these are the optimal s-stage methods of order $2s$) and methods that can be constructed from GLRK methods by composition conserve all quadratic first integrals.

The simplest GLRK method is the implicit midpoint rule. To see that it conserves quadratic invariants, suppose $C(x) = \frac{1}{2} x^T A x$, with $A = A^T$, is a first integral such that the condition $x^T A f(x) = 0$, for all $x \in \mathcal{D}$. Then multiplying both sides of (4) by $\frac{1}{2}(x_{n+1} + x_n)^T A$ gives

$$\left(\frac{x_{n+1} + x_n}{2}\right)^T A \left(\frac{x_{n+1} - x_n}{\Delta t}\right) = \left(\frac{x_{n+1} + x_n}{2}\right)^T A f \left(\frac{x_{n+1} + x_n}{2}\right) = 0.$$

The above simplifies to

$$x_{n+1}^T A x_{n+1} - x_n^T A x_n = C(x_{n+1}) - C(x_n) = 0.$$

There are no Runge-Kutta methods that conserve all polynomial first integrals of degree p for $p > 2$.

Specialist methods exist for conserving first integrals for problems where these are explicitly known. In particular, we mention *discrete gradient methods* in this context.

5 Symplectic numerical integrators for Hamiltonian systems

A symplectic numerical method is one whose map is a symplectic map. In other words, a one step method

$$x_{n+1} = \psi_{\Delta t}(x_n)$$

is symplectic (with respect to J) if

$$D\psi_{\Delta t}^T(x) J^{-1} D\psi_{\Delta t}(x) = J^{-1}, \quad \forall x \in \mathcal{D}.$$

It turns out that the class of Gauss-Legendre Runge-Kutta methods is symplectic for any J. (This result is a consequence of the conservation of quadratic first integrals, combined with the fact the discretization and linearization commute for Runge-Kutta methods). In particular the implicit midpoint rule is a symplectic method:

$$\frac{x_{n+1} - x_n}{\Delta t} = J\nabla H\left(\frac{x_{n+1} + x_n}{2}\right).$$

The *canonical* symplectic form on $\mathbf{R}^{2d}$ is

$$J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix},$$

where I is the identity matrix of dimension d. For any Hamiltonian system with non-singular J, there exists a linear transformation to a Hamiltonian system with canonical symplectic form. In the canonical case, we write

$$x = \begin{pmatrix} q \\ p \end{pmatrix},$$

and (3) can be written

$$\dot{q} = \nabla_p H(q,p), \qquad \dot{p} = -\nabla_q H(q,p).$$

For the canonical case, some other symplectic methods are the *symplectic Euler methods*

$$\begin{aligned} q_{n+1} &= q_n + \Delta t \nabla_p H(q_{n+1}, p_n) \\ p_{n+1} &= p_n - \Delta t \nabla_q H(q_{n+1}, p_n), \end{aligned}$$

and (its adjoint)

$$\begin{aligned} q_{n+1} &= q_n + \Delta t \nabla_p H(q_n, p_{n+1}) \\ p_{n+1} &= p_n - \Delta t \nabla_q H(q_n, p_{n+1}). \end{aligned}$$

These methods are first order methods, but they can be composed with each other (in either order) to give second order self-adjoint methods.

In many physics problems, the Hamiltonian takes the seperable form $H(q,p) = K(p) + V(q)$ of the sum of kinetic and potential energies. For these problems, the methods above become explicit, and the second order composition can be written

$$\begin{aligned} q_{n+1/2} &= q_n + \frac{\Delta t}{2} \nabla_p K(p_n) \\ p_{n+1} &= p_n - \Delta t \nabla_q V(q_{n+1/2}) \\ q_{n+1} &= q_{n+1/2} + \frac{\Delta t}{2} \nabla_p K(p_{n+1}), \end{aligned}$$

(and the counterpart with opposite composition order) which is referred to as the Störmer-Verlet method. The method is explicit, self-adjoint and symplectic. It is the work-horse of molecular dynamics simulations in computational chemistry, as well as in gravitational simulations in celestial mechanics.

One can also construct symplectic methods using splitting. Suppose $F, G : \mathcal{D} \to \mathcal{D}$ are symplectic maps, then their composition is also a symplectic map. Since $D(F \circ G)(x) = DF(G(x))DG(x)$,

$$D(F \circ G)^T J^{-1} D(F \circ G) = DG^T\, DF^T J^{-1} DF\, DG = DG^T J^{-1} DG = J^{-1}.$$

Since the exact flow of a Hamiltonian system is a symplectic map, we can construct symplectic splitting methods by splitting the Hamiltonian and solving the individual Hamiltonian flows exactly. Suppose $H(x) = H_1(x) + H_2(x)$ where the individual flows $\phi_{1,t}(x)$ and $\phi_{2,t}(x)$ are the respective exact flows of the systems

$$\phi_{1,t}(x) : \dot{x} = J\nabla H_1(x), \qquad \phi_{2,t}(x) : \dot{x} = J\nabla H_2(x)$$

and can be exactly constructed. Then the composition

$$\psi_{\Delta t} = \phi_{1,\Delta t} \circ \phi_{2,\Delta t}$$

is a first order symplectic method, as is its adjoint $\psi^*_{\Delta t}$. And of course, the composition of $\psi_{\Delta t/2} \circ \psi^*_{\Delta t/2}$ (and its reverse composition) are second order symplectic methods. In particular, in the canonical case with separable Hamiltonian $H(q,p) = K(p) + V(q)$, the splitting $H_1 = K(p)$, $H_2 = V(q)$ yields the Störmer-Verlet method above.

Since symplectic methods are symplectic maps, they preserve the Lebesgue measure, which means that if we consider a bounded open set of initial conditions, the volume of the set is preserved under symplectic maps. This fact turns out to be important for the statistics of long simulations (i.e. measurable sets preserve their measure).

However, more can be said about symplectic methods using *backward error analysis*. Given a numerical method $\psi_{\Delta t}$ applied to (1), we can construct a new vector field via an asymptotic expansion,

$$\tilde{f}(x) = f(x) + \Delta t f_1(x) + \Delta t^2 f_2(x) + \cdots,$$

and demand that the Taylor series expansion of the exact flow $\phi_{\Delta t}$ of the modified differential equation $\frac{d}{dt}\tilde{x} = \tilde{f}(\tilde{x})$ agrees with the map $\psi_{\Delta t}$ (i.e. we use this condition to determine the vector fields $f_1,\ f_2,\ \ldots$). The truncated asymptotic expansions tell us something about the behavior of the numerical method, since they agree with the method to higher order accuracy than the original problem.

For instance, for the forward Euler method, it is easy to check that $f_1 = -\frac{1}{2}Df(x)f(x)$, so while the truncation error of forward Euler applied to (1) is $\mathcal{O}(\Delta t^2)$, the computed value is actually an $\mathcal{O}(\Delta t^3)$ approximation of the differential equation

$$\frac{d}{dt}\tilde{x} = f(\tilde{x}) - \frac{\Delta t}{2} Df(\tilde{x})f(\tilde{x}).$$

For Hamiltonian systems, it turns out that if we use a symplectic method, the modified vector field has the special form

$$\tilde{f}(x) = J\nabla\tilde{H}(x), \qquad \tilde{H}(x) = H(x) + \Delta t H_1(x) + \Delta t^2 H_2(x) + \cdots .$$

In other words, the modified differential equation is again a Hamiltonian system! That means the modified Hamiltonian $\tilde{H}$ is a first integral of the numerical method, i.e. $\tilde{H} \circ \psi_{\Delta t}(x) = \tilde{H}(x)$. Consequently, the original Hamiltonian satisfies

$$H(x) = \tilde{H}(x) - \Delta t H_1 - \Delta_2^2 H_2 + \cdots$$

If the numerical method is of (global error) order p, then $H_1 = \cdots = H_{p-1} = 0$. In practice, if the Hamiltonian is sufficiently smooth, we typically see

$$|H(x_n) - \tilde{H}(x_0)| < C\Delta t^p,$$

which means that the Hamiltonian is conserved up to bounded fluctuations of $\mathcal{O}(\Delta t^p)$.

Together with the aforementioned conservation of volume, this result makes symplectic methods very powerful for long simulations of complex Hamiltonian systems, where computing statistics is often the goal.